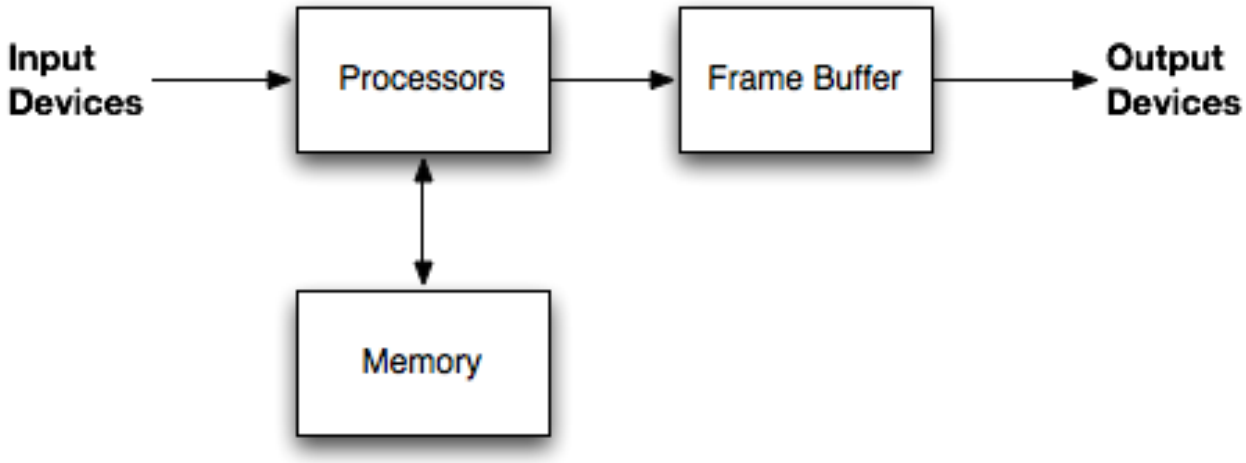
SEC 1

The basic components of computer graphics system

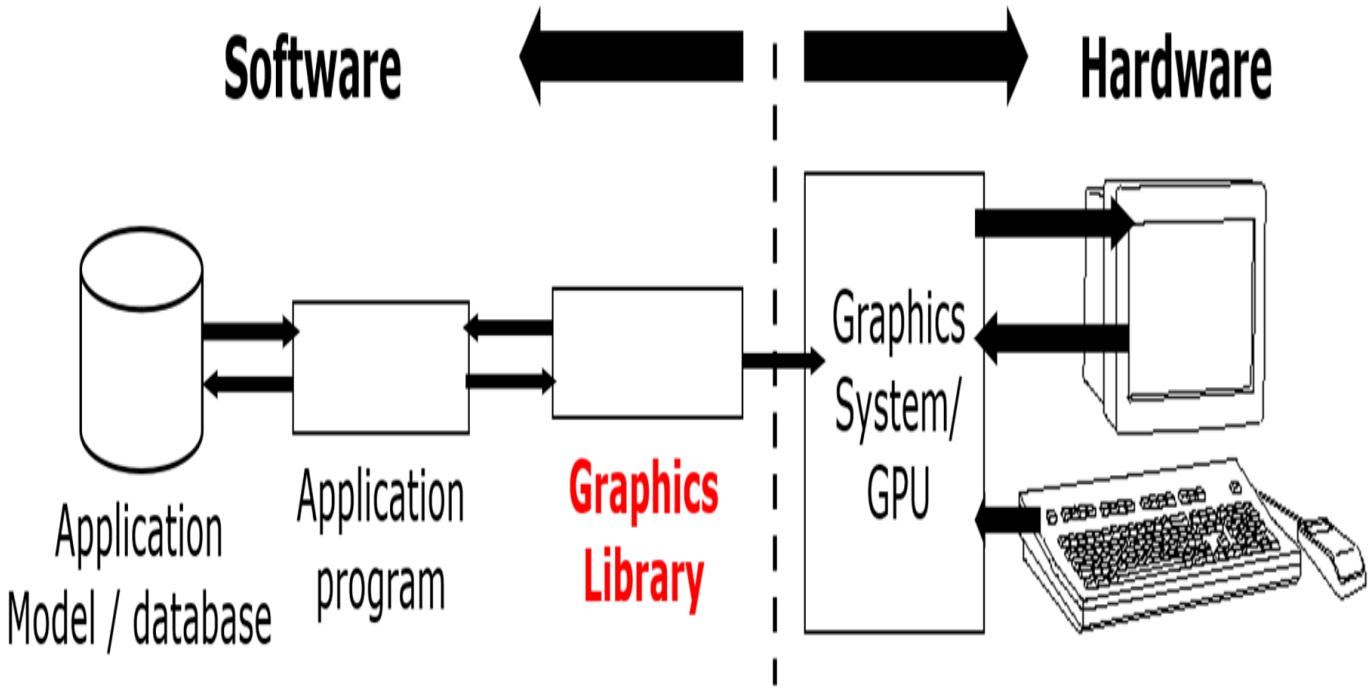


- Different types of data are retrieved from input devices called input primitives such as:
  - Locator Devices such as mouse → Input Primitive: Coordinate position (x,y)
  - Keyboard →Input Primitive: A string of characters
  - Valuator→ Input primitive: Scalar values (typically between 0 and 1)

- O/P device displays data from memory of the computer. Examples: CRT display, LED display, LCD display, Plasma display, Plotter, hologram generator.
  - 2-D output devices are either vector devices, which physically draw line segments (like a plotter), or raster devices, which display a grid of colored pixels (picture elements).
- Processor (CPU &GPU)
  - CPU gets the specifications for a geometric shape from input device and hands the geometric data with additional operations) to the GPU
  - The GPU rasterizes the data by deciding which pixels to use to represent the data then places the pixel data into the frame buffer to show on the display screen
- Frame buffer: It is a block of memory dedicated to graphics output which holds data representing all the pixels in a complete video frame.

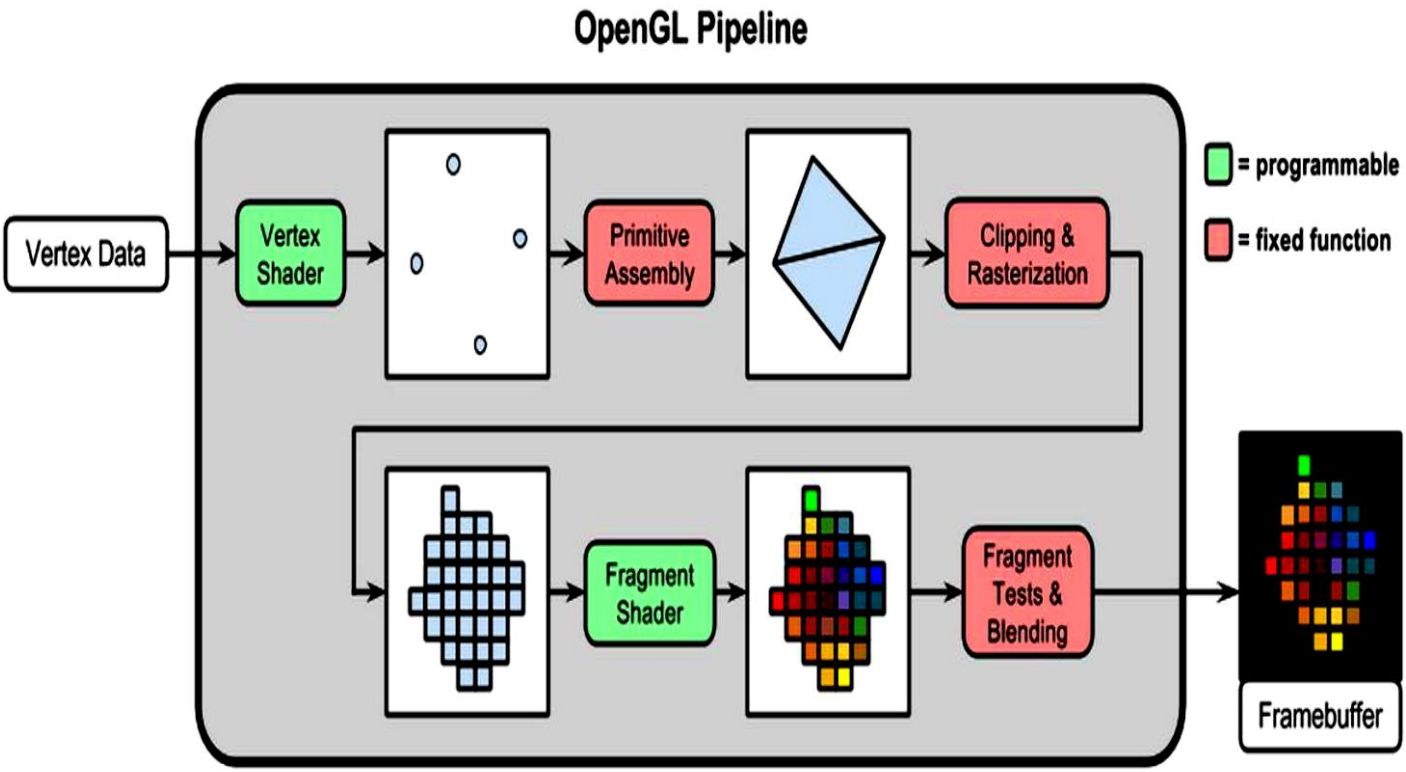
Sec 2

1. Conceptual Framework of Computer Graphics "how a graphics system is organized, what pieces exist and how they fit together"



- The computer graphic system Conceptual Framework consists of two major components at two levels:
  - Hardware level: the computer receives input from interactive devices and show images on display devices.
  - Software level: it involves three components
    - Application model→ it captures all the data or objects to be pictured on the screen that stored in application Database.
    - Application program →creates, store into and retrieved from application model such as word, excel, paint, or Photoshop program (handles user's input)
    - Graphics library or application programmer's interface (API)→ is a set of procedures for handling graphics Such as OpenGL, Direct3D or Java3D. it is intermediary between application and display hardware (Graphics System).

2. OpenGL pipeline.



Graphics pipeline is a conceptual model that describes what steps a graphics system needs to perform to render a 3D scene to a 2D screen

- The vertex shader → define object vertices to move object around
- Clipping → cutting vertices that will not on the screen
- Rasterization → creating pixels for each part of the object
- Fragment shader → add values for each pixel (coloring object shape)

3. The distinction between:

- Computer graphics and image processing.
  - computer graphics are concerned with the creation of images
  - Image Processing: analyze pictures to derive descriptions of objects appeared in the pictures.

- Geometric and mathematical modeling.
    - Geometric model : describes how objects appears (it defines attributes that describe color texture or shape of the object)
    - Mathematical model: is a model of a physical process in which it describes how object moves
- Non-interactive and interactive graphics systems.
    - Non-interactive computer graphics: It involves only one-way communication between the computer and the user where users can see the produced image, and cannot make any change.
    - Interactive computer graphics: It involves two-way communication between the computer and the user as shown in games.
- vector and raster display technologies
    - Raster display technologies it composed of pixels that provide more details shape
    - Vector display technologies it composed of connected sequence of points (paths) that provide smooth shape but limited to set of colors which used for less details Images such as logos